A brief introduction to Brownian motion

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Gaussian processes

Definition 1. *A continuous-time stochastic process $\{X(t)\colon t \geqslant 0\}$ is defined as a Gaussian (or normal) process if for any integer $n \geqslant 1$ and any collection of times $t_1, \dots, t_n$, the random variables $X(t_1), \dots, X(t_n)$ jointly follow a multivariate normal distribution.*

The mean vector and covariance matrix fully characterise a multivariate normal distribution. Therefore, the statistical properties of a Gaussian process are specified by its mean value function, $m_X(t)$, and its autocovariance function, $C_X(s,t)$. For example, since $\mathsf{Var}(X(t)) = C_X(t,t)$, the first-order density function of a Gaussian process $X(t)$ is given by:

$$f_X(x;t) = \frac{1}{\sqrt{2\pi\, C_X(t,t)}} \exp\left(-\frac{(x - m_X(t))^2}{2\, C_X(t,t)}\right). \tag{1}$$

The general form of the n-order density function can be expressed as follows:

$$f_X(x_1, \dots, x_n; t_1, \dots, t_n) = \frac{1}{2\pi\sqrt{\det(\boldsymbol{K})}} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{m}_X)^t \boldsymbol{K}^{-1}(\boldsymbol{x} - \boldsymbol{m}_X)\right),$$

where $\boldsymbol{x} = (x_1, \dots, x_n)^t$, $\boldsymbol{m}_X = (\mathbb{E}(X(t_1), \dots \mathbb{E}(X(t_n)))^t = (m_X(t_1), \dots, m_X(t_n))^t$, and the elements of the covariance matrix $\boldsymbol{K} = (k_{ij})$ are:

$$\begin{aligned} k_{ii} &= \mathsf{Var}(X(t_i)) = C_X(t_i, t_i), \\ k_{ij} &= \mathsf{Cov}(X(t_i), X(t_j)) = C_X(t_i, t_j), \quad i \neq j. \end{aligned}$$

Because $m_X(t)$ and $C_X(s,t)$ completely determine the probabilistic properties of a Gaussian process, the following important result is established:

Theorem 1. *If a Gaussian process is wide-sense stationary, then it is also strictly stationary.*

Brownian motion as a limit of the random walk.

Consider the symmetric random walk $\{S_n\colon n \geqslant 0\}$. For each time increment Δt, a step of size Δx is taken. A new discrete-time, discrete-state stochastic process, denoted $X[n]$, is defined as follows:

$$X[n] \equiv X(n\Delta t) = \Delta x \cdot S_n, \quad n \geqslant 0.$$

As is customary, we assume that $S_0 = 0$, which implies that $X[0] = 0$.

Given that $\mathbb{E}(S_n) = 0$ and $\mathsf{Var}(S_n) = n$, it follows that $\mathbb{E}(X[n]) = 0$ and $\mathsf{Var}(X[n]) = n(\Delta x)^2$. According to the Central Limit Theorem, the following result is obtained:

$$\frac{X[n]}{\sqrt{n}\,\Delta x} \xrightarrow{d} \mathsf{N}(0,1) \quad \text{as } n \to \infty.$$

Let $t > 0$ represent a positive time. For a given $n \geqslant 1$, define $\Delta t = t/n$ and set

$$\Delta x = \sigma\sqrt{\Delta t} = \sigma\sqrt{\frac{t}{n}}$$

where σ is a positive constant. The variance of $X[n]$ can be expressed as

$$\mathrm{Var}(X[n]) = n(\Delta x)^2 = \sigma^2 t.$$

In the limit as $n \to \infty$, both Δt and Δx approach zero. This limiting process yields a continuous-time and continuous-state random process defined as

$$X(t) \equiv \lim_{n\to\infty} X[n],$$

such that:

- For each $t > 0$, the random variable $X(t)$ follows a Gaussian $\mathsf{N}(0, \sigma^2 t)$ distribution.
- Because the changes in the value of the random walk S_n in non-overlapping time intervals are independent, the process $\{X(t)\colon t \geqslant 0\}$ has what are known as *independent increments*. This means that for any sequence of times $t_1 < t_2 < \cdots < t_{n-1} < t_n$, the random variables $X(t_1)$, $X(t_2) - X(t_1)$, $X(t_3) - X(t_2)$, ..., and $X(t_n) - X(t_{n-1})$ are all independent from one another.
- Since $\mathbb{P}(S_m - S_n = s) = \mathbb{P}(S_{m-n} = s)$, the stochastic process $\{X(t)\colon t \geqslant 0\}$ possesses *stationary increments*, meaning the distribution of $X(t+s) - X(t)$ is independent of the instant t and equals the distribution of $X(s)$.

These properties are formalised in the following definition.

Definition 2. *A continuous-time stochastic process $\{X(t)\colon t \geqslant 0\}$ is a Brownian motion if:*

1. *$X(0) = 0$.*
2. *$\{X(t)\colon t \geqslant 0\}$ has stationary and independent increments.*
3. *For each $t > 0$, $X(t)$ is normally distributed with mean 0 and variance $\sigma^2 t$.*

Brownian motion, also referred to as the Wiener process, is interpreted as the limit of the symmetric random walk. This interpretation suggests that each realisation of the process is a continuous function of time, and, indeed, it can be demonstrated that $X(t)$ is continuous with respect to t with probability one. However, $X(t)$ is nowhere differentiable, which is a defining property of this process. The inherent randomness of Brownian motion makes it unsuitable for integration using conventional methods.

If $\sigma = 1$, the process is known as standard Brownian motion. Any Brownian motion can be transformed into a standard one by setting $B(t) = X(t)/\sigma$. Therefore, we will assume throughout this discussion that $\sigma = 1$.

Let us determine the n-th order density of the process. Firstly, observe that the first-order density of $X(t)$ is given by

$$f(x;t) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}, \quad -\infty < x < \infty. \tag{2}$$

According to the assumption of independent increments, the variables $Y_1 \equiv X(t_1)$, $Y_2 \equiv X(t_2) - X(t_1)$, …, and $Y_n \equiv X(t_n) - X(t_{n-1})$ are independent of each other. Additionally, based on the assumption of stationary increments, each Y_k follows the distribution of $X(t_k - t_{k-1})$, which is normally distributed with a mean equal to zero and a variance equal to $t_k - t_{k-1}$. Therefore, by applying the change of variable theorem, we can determine the joint density of the variables

$$X(t_1) = Y_1, \quad X(t_2) = Y_1 + Y_2, \quad \ldots, \quad X(t_n) = Y_1 + Y_2 + \cdots + Y_n,$$

which is given by:

$$
\begin{aligned}
f(x_1, x_2, \ldots, x_n; t_1, t_2, \ldots, t_n) &= f(x_1; t_1) f(x_2 - x_1; t_2 - t_1) \cdots f(x_n - x_{n-1}, t_n - t_{n-1}) \\
&= \frac{1}{(2\pi)^{n/2}\sqrt{t_1(t_2 - t_1)\cdots(t_n - t_{n-1})}} \exp\left[-\frac{1}{2}\left(\frac{x_1^2}{t_1} + \frac{(x_2 - x_1)^2}{t_2 - t_1} + \cdots + \frac{(x_n - x_{n-1})^2}{t_n - t_{n-1}}\right)\right]. \qquad (3)
\end{aligned}
$$

From this equation, we can theoretically calculate any desired probabilities.

Each random variable $X(t_1), X(t_2), \ldots, X(t_n)$ can be represented as a linear combination of the independent normal random variables $Y_1, Y_2, \ldots, Y_n$. Consequently, Brownian motion can also be defined as a Gaussian process with $\mathbb{E}(X(t)) = 0$, and such that for $s < t$, the autocovariance function can be expressed as:

$$
\begin{aligned}
C_X(s,t) &= \mathsf{Cov}(X(s), X(t)) = \mathsf{Cov}(X(s), X(s) + (X(t) - X(s)) \\
&= \mathsf{Var}(X(s)) + \mathsf{Cov}(X(s), X(t) - X(s)) = \mathsf{Var}(X(s)) = s,
\end{aligned}
$$

where we note that $\mathsf{Cov}(X(s), X(t) - X(s)) = 0$ because $X(s)$ and $X(t) - X(s)$ are independent.

Remark 1. Notice that for the standard Brownian motion, expressions (1) and (2) coincide because $C_X(t,t) = t$. Furthermore, if $t_1 < t_2$, the covariance matrix of $X(t_1)$ and $X(t_2)$ is

$$
K = \begin{pmatrix} t_1 & t_1 \\ t_1 & t_2 \end{pmatrix},
$$

which inverse is

$$
K^{-1} = \frac{1}{t_1(t_2 - t_1)} \begin{pmatrix} t_2 & -t_1 \\ -t_1 & t_2 \end{pmatrix}
$$

so that

$$
\begin{aligned}
(x_1 \ x_2)\, K^{-1}\, (x_1 \ x_2)^t &= \frac{x_1^2 t_2 - 2x_1 x_2 t_1 + x_2^2 t_1}{t_1(t_2 - t_1)} \\
&= \frac{t_1(x_1 - x_2)^2 + x_1^2(t_2 - t_1)}{t_1(t_2 - t_1)} = \frac{x_1^2}{t_1} + \frac{(x_1 - x_2)^2}{t_2 - t_1}.
\end{aligned}
$$

Hence, the second-order density,

$$
f(x_1, x_2; t_1, t_2) = \frac{1}{2\pi\sqrt{\det(K)}} \exp\left(-\frac{1}{2}(x_1 \ x_2)\, K^{-1}(x_1 \ x_2)^t\right),
$$

is given by expression (3) for $n = 2$.

Hitting times

Let T_a represent the first time when the Brownian motion reaches the point a on the coordinate axis.

If $a > 0$, we can compute $\mathbb{P}(T_a \leqslant t)$ by considering $\mathbb{P}(X(t) \geqslant a)$. By conditioning on whether or not $T_a \leqslant t$, we have:

$$
\mathbb{P}(X(t) \geqslant a) = \mathbb{P}(X(t) \geqslant a \mid T_a \leqslant t)\, \mathbb{P}(T_a \leqslant t) + \mathbb{P}(X(t) \geqslant a \mid T_a > t)\, \mathbb{P}(T_a > t). \qquad (4)
$$

Let us see that expression (4) simplifies to:

$$
\mathbb{P}(X(t) \geqslant a) = \frac{1}{2}\, \mathbb{P}(T_a \leqslant t).
$$

Indeed, by symmetry, we have

$$
\mathbb{P}(X(t) \geqslant a \mid T_a \leqslant t) = \mathbb{P}(X(t) \leqslant a \mid T_a \leqslant t) = \frac{1}{2},
$$

and, additionally,

$$\mathbb{P}(X(t) \geqslant a \mid T_a > t) = 0$$

because, due to continuity, the process cannot exceed a without having first reached a.

For $a < 0$, the distribution of T_a is, by symmetry, the same as that of T_{-a}.

Therefore

$$\mathbb{P}(T_a \leqslant t) = 2\,\mathbb{P}(X(t) \geqslant |a|) = \frac{2}{\sqrt{2\pi t}} \int_{|a|}^{\infty} e^{-x^2/(2t)}\, dt = \frac{2}{\sqrt{2\pi}} \int_{|a|/\sqrt{t}}^{\infty} e^{-u^2/2}\, du. \tag{5}$$

Another random variable of interest is the maximum value the process achieves in the interval $[0, t]$. For $a > 0$, its distribution is also described by (5), since, by continuity,

$$\mathbb{P}\left(\max_{0 \leqslant s \leqslant t} X(s) \geqslant a\right) = \mathbb{P}(T_a \leqslant t).$$

Brownian motion with drift and geometric Brownian motion

Definition 3. *A continuous-time stochastic process $\{X(t)\colon t \geqslant 0\}$ is a Brownian motion with drift parameter μ and variance parameter σ^2 if:*

1. *$X(0) = 0$.*
2. *$\{X(t)\colon t \geqslant 0\}$ has stationary and independent increments.*
3. *For each $t > 0$, $X(t)$ is normally distributed with mean μt and variance $\sigma^2 t$.*

An equivalent definition is to let $\{B(t)\colon t \geqslant 0\}$ be standard Brownian motion and then define

$$X(t) = \sigma B(t) + \mu t.$$

Definition 4. *Let $\{Y(t)\colon t \geqslant 0\}$ be a Brownian motion with drift parameter μ and variance parameter σ^2. The stochastic process $\{X(t)\colon t \geqslant 0\}$ defined by*

$$X(t) = e^{Y(t)}$$

is called a geometric Brownian motion.

Let $\{X(t)\colon t \geqslant 0\}$ be a geometric Brownian motion. Then, for $s < t$,

$$\begin{aligned}
\mathbb{E}(X(t) \,|\, X(u)\colon 0 \leqslant u \leqslant s) &= \mathbb{E}\left(e^{Y(t)} \,|\, Y(u)\colon 0 \leqslant u \leqslant s\right) \\
&= \mathbb{E}\left(e^{Y(s)+(Y(t)-Y(s))} \,|\, Y(u)\colon 0 \leqslant u \leqslant s\right) \\
&= e^{Y(s)}\, \mathbb{E}\left(e^{Y(t)-Y(s)} \,|\, Y(u)\colon 0 \leqslant u \leqslant s\right) = e^{Y(s)}\, \mathbb{E}\left(e^{Y(t)-Y(s)}\right).
\end{aligned}$$

The penultimate equality holds because $Y(s)$ is determined by the values of $Y(u)$ for $0 \leqslant u \leqslant s$. The final equality is justified by the independent increments property of Brownian motion.

Moreover, since the moment generating function of a normal random variable W is

$$\phi_W(z) = \mathbb{E}\left(e^{zW}\right) = e^{m_W z + (1/2)\sigma_W^2 z^2},$$

and since $Y(t) - Y(s)$ is normal with expectation $\mu(t - s)$ and variance $\sigma^2(t - s)$, we obtain (by setting $z = 1$) that

$$\mathbb{E}(X(t) \,|\, X(u)\colon 0 \leqslant u \leqslant s) = X(s)\, e^{(t-s)(\mu+\sigma^2/2)}.$$

Geometric Brownian motion serves as a widely used stochastic process for modeling stock prices over time, especially under the assumption that percentage changes are independent and identically distributed. For example, suppose that X_n is the price of a particular stock at time n and assume that $Y_n = X_n/X_{n-1}$, $n \geqslant 1$, are independent and identically distributed random variables. Therefore,

$$X_n = Y_n X_{n-1} = Y_n Y_{n-1} X_{n-2} = \cdots = Y_n Y_{n-1} \cdots Y_1 X_0.$$

Thus,

$$\log(X_n) = \sum_{i=1}^{n} \log(Y_i) + \log(X_0).$$

Given that the random variables $\log(Y_i)$ are independent and identically distributed for $i \geqslant 1$, the sequence $\{\log(X_n) \colon n \geqslant 1\}$, when appropriately normalised, exhibits behaviour similar to a Brownian motion with drift. Consequently, the sequence $\{X_n \colon n \geqslant 1\}$ can be approximated by a geometric Brownian motion.